
Study on budgetary gap analysis of Diarrhoea and Pneumonia commodities at provincial/region level

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Abbreviations

ARI	Acute Respiratory Infection
MDG	Millennium Development Goals
PDHS	Pakistan Health and Demographic Survey
MICS	Multiple Index Cluster Survey
PBS	Pakistan Bureau of Statistics
PDHS	Pakistan Demographic and Health Survey
KP	Khyber Pakhtunkhwa
FATA	Federally Administered Tribal Areas
PSLM	Pakistan Social & Living Standard Measurement
DDO	Drawing and Disbursing Officer
WHO	World Health Organisation
DHO	District Health Officer
DHIS	District Health Information System
CEO	Chief Executive Officer
LHW	Lady Health Worker
LHS	Lady Health Supervisor
MNCH	Maternal Neonatal and Child Health
OH	Overheads
CMW	Community Midwives
MPA	Member of Provincial Assembly
MNA	Member of National Assembly

1. Executive Summary

Pakistan¹ ranks in the countries with high infant mortality rates. As per the available statistics, each year around 89 children under the age of 5 die in 1,000 livebirths. While there are different causes of high mortality, the two most noticeable reasons are acute respiratory infection (pneumonia) and diarrhoea. It was reported² in 2012-13 that 91,000 children died from pneumonia and 53,300 children died from diarrhoea.

As per the Pakistan Demographic and Health Survey (PDHS) of 2012-13, around 23% of children under the age of 5 had diarrhoea two weeks before the survey, 61% were taken to appropriate health provider, only 38% were provided with an oral rehydration salts (ORS), and less than 2% were provided with zinc supplements. Similarly, 16% of children under the age of 5 had acute respiratory infection (ARI) / pneumonia two weeks before the survey, and 42% with ARI received antibiotics.

This study estimates the funding required to reduce child mortality and morbidity caused due to diarrhoea and pneumonia. Study objectives and the methodology adopted is provided in Chapter 2. Specific study findings are provided in Chapter 3 while recommendations are provided in Chapter 3.8.

To undertake this study, visits were made to 44 health facilities and 5 warehouses in 10 districts of Punjab and Sindh. In addition, data collection exercise was undertaken through the provincial headquarters.

Key study findings include:

1. Only recently provisional³ population census numbers have been announced by the Pakistan Bureau of Statistics. As per the estimated Pakistan has a total population of 207.8 million. While demographic statistics of age cohort have not yet been published, using the age cohort of 1998, it is estimated that there are 27.8 million children under the age of 5 years (Table 3).
2. Based on the above population figures and number of morbid cases reported by Pakistan Health and Demographic Survey (PDHS) of 2012-13, each year the number of children with diarrhoea are estimated to be 6.3 million (Table 4) and children with ARI are estimated to be 4.4 million (Table 6).
3. Based on PDHS 2012-13's estimate of 4 to 6 episodes of diarrhoea and 1 to 2 episodes of pneumonia (ARI - Acute Respiratory Infection) each year, the total amount required to treat all the morbid under 5 population of children is estimated to be around Rs.2.3 billion (around \$210 million) - Table 13.
4. At the level of community health workers, primary and secondary healthcare services, the government is estimated to have spent Rs.830 million rupees per year (which includes cost of medicines, labour and overhead cost (Table 12) in 2015-16. This means

¹ 1 UNICEF. (2015). Committing to Child Survival: A Promise Renewed. Progress Report 2015 - http://www.unicef.org/publications/files/APR_2015_9_Sep_15.pdf

² PDHS 2012-13

³ <http://www.pbs.gov.pk/node/2047> - The provisional census data is being challenged by different political parties and it is not clear if these numbers will remain the same in the future

that the budgetary gap between what is required and what is supplied is a little under 3 times.

5. The government procures ORS, Zinc suspension and Amoxicillin syrup at a considerably cheaper rate than what is available in the market (Table 10). Therefore, the out of pocket expenditure - around 64% of total health expenditure as per National Health Accounts 2005-06 - is estimated at a high level of Rs.3.1 billion (Table 14).
6. In a large number of health facilities visited, ORS, Zinc and Amoxicillin was in stock. There were however, a few health facilities where Zinc was out of stock. It was noted that many lady health workers were out of stock for the past few months.
7. Provinces have some differences in the mode of provision of health services. Sindh has moved towards outsourcing - public private partnership (PPP) - modal. Procurements are undertaken by these outsourced organisations within single-line budgets. Punjab has formed autonomous organisations (district health authorities, and health facilities management company) and discontinued the PPP mode. Operations of these autonomous organisations, including recruitments and definition of rules and regulations was under review at the time of undertaking this study.
8. Budgetary management practices need improvement especially related to budget planning, formulation, and funds releases.

Based on this study the following key recommendations are made:

1. Improvement in the method of Forecasting Procurement of Medicines

Currently the forecasting of procurements is based on consumption. Gradually, this system needs to be modified to procurements based on multiple criteria including consumption, morbidity, and out of pocket expenditures.

2. An early completion of Central Rate Contracts

Central rate contracts in the provinces should be finalised by July or by maximum by August each year. All Drawing and Disbursing Officers (DDOs) should be notified on timely basis (preferably by August end each year) of the budget available to them, and the finalisation of central rate contracts and procurement guidelines.

3. Procurement and distribution system in case of vertical health programmes

Emphasis will need to be placed on sufficient and timely availability of medicines for diarrhoea and pneumonia treatment with the lady health workers. Going forward, as the Federal Government is planning to reduce its funding for vertical health programmes, a way forward plan on financing by provinces will need to be formulated.

4. An increase of budget on procurement of medicines for treatment of diarrhoea and pneumonia

In Balochistan, KP and FATA, there is an immediate need to enhance budget allocation for medicines especially for treatment of diarrhoea and pneumonia as discussions with provincial departments of health revealed that stock outs occur. Special emphasis will need to be placed on rural areas. In Sindh and Punjab, while supply levels of diarrhoea and pneumonia commodities have increased over the past 2 - 3 years, there is a further need to enhance stock levels by 10 - 15%.

5. Improvement in budget management practices

Discussions will need to take place with Finance Departments to allow one-time release of the budget head 'procurement of drugs and medicines'. Currently the release policy allows quarter funds released every quarter. It was noted during discussions with provincial and district authorities that releases are often delayed resulting in delayed procurement at the sub-provincial level. In addition, every DDO should be notified of his/her budget amounts by 30 August each year. Trainings should be provided to DDOs on procurement. In Punjab rules and regulations and appropriate budgets should be allowed on timely basis for district health authorities and facilities management company. In Sindh, funds release issues should be resolved with the IHS.

6. Development of IT systems

Centralised stocks register - including data of procurements, stocks in use, and stock outs, need to be maintained at the central level (departments of health) to enhance efficiency of use and allocations. In addition, the IT systems that have been developed by Punjab government may be demonstrated to other provinces with the aim to replicate such systems.

7. Determination of policy on reduction of out-of-pocket expenditure

Out of pocket expenditure in Pakistan is high⁴. At times, it results in catastrophic expenditure. If the government intends to provide universal healthcare services, then a policy needs to be formulated on how the out of pocket expenditure will be reduced. This can be achieved through devising a more holistic initiative of community health workers, provision of health insurance, increase in government's budgetary allocations, etc.

8. Holding of pre-budget consultative sessions / seminars with different stakeholders

It is recommended that for budget 2018-19, for which the budget preparation activities, in the provinces will start from November / December this year, pre-budget consultative workshops should be held with relevant provincial and district officials, local level politicians, and donors. The aim of these workshops will be to apprise various stakeholders of the demographic, disease burden and current levels of health services in the districts and local communities.

9. Promotion of the use of co-packing (ORS and Zinc tablets) for treatment of diarrhoea and Amoxicillin suspension tablets for treatment of pneumonia in the private sector

Private sector role in treatment of diarrhoea and pneumonia is significant and therefore, the government should, based on WHO's recommendation, promote use of co-packing (ORS and Zinc DT), and Amoxicillin suspension tablets with general physicians.

Details on the recommendations are provided in Chapter 3.8.

⁴ Source; National Health Survey 2005-06

2. Study Objectives and Methodology

2.1. Study Objectives

This study assesses financial resources required vs available at the provincial and area levels for the treatment of Pneumonia and Diarrhoea by the public sector in Pakistan. More specifically, this study:

- Estimates the current level of funding commitment at provincial/regional (i.e. sub-provincial) levels by Government towards purchase of commodities for treatment of Pneumonia and Diarrhoea by using the information provided by the Government.
- Provides a forecasting mechanism to ascertain funding requirements to purchase essential commodities for treatment of Pneumonia and Diarrhoea at the primary and secondary level.
- Uses certain methods to assess the direct costs incurred by the Government (i.e. cost of procurement and cost of transportation) and indirect costs (i.e. manpower cost, overheads, etc.) incurred by the Government.
- Provides a method of calculating the total amount of budget demanded vs allocated and spent on essential commodities for the treatment of Pneumonia and Diarrhoea.
- Discusses the ways to reduce out of pocket expenditure in Pakistan through the review of health treatment offered by the private health facilities vs. the public sector.
- Determines the diversity and depth of the budget performance and implementation challenges.
- Provide recommendations for adequate resource allocation and methods to improve forecasting at the provincial and sub-provincial levels.

2.2. Scope of the Study

The study was carried out for Balochistan, FATA, Federal Government, Khyber Pakhtunkhwa, Punjab and Sindh. This study does not include analysis for Azad Jammu and Kashmir, and Gilgit Baltistan.

In the above geographical locations, the study focuses on public sector health provision for the treatment of diarrhoea and pneumonia in children under the age of 5. More specifically the study included costs incurred on treatment of the two diseases by the civil government including:

1. Departments of health and district governments (district health offices / chief executive officers),
2. Vertical health programmes (lady health workers programme, community midwives' programmes, and nutrition programme),
3. Special programmes run by the government (e.g. control of diarrhoea disease programme in Sindh),
4. Autonomous government hospitals, and
5. Outsourced health provision at the primary level under public-private-partnership model.

The study excludes costs of treatment incurred by the military through the government's defence budget. The military on average spends around 4%⁵ of total health expenditure in the country.

The study used data of 132 districts across Pakistan - which includes 6 regions and 7 agencies in FATA. The data was largely collected from provincial authorities.

To review health treatment of the two diseases offered by the private health facilities, discussions were held with 3 private clinics in Punjab.

2.3. Key Informant Interviews

To undertake this study, key informant interviews were held with health facilities, warehouses and lady health workers in 6 districts of Sindh and 4 districts of Punjab. A total of 44 health facilities and 5 warehouses were visited.

Table 1: Health Facilities Visited

	Sindh	Punjab
DHO/Civil Hospitals	4	4
THQ Hospitals	5	3
RHCs*	6	4
BHUs**	11	7

*Including RHCs managed by IHS and BHUs managed by PPHI in Sindh

The aim of the visits was to:

- Study the system of forecasting of commodities required for treatment of diarrhoea and pneumonia,
- Study existing prescription practices,
- Document the budgetary management system in the health facilities - especially budget demands, budget releases, and reporting mechanisms, and
- Understand the system of procurement, management of medicine stock outs, and treatment guidelines and practices.

Key informant interviews were also held with provincial departments of health (including vertical health programmes), departments of finance, departments of planning and development and outsourced entities (e.g. PPHI and IHS in Sindh) to understand methods of budget making, funds releases and expenditure management.

In addition, discussions were also held with doctors in private clinics to understand treatment offered by the private sector to pneumonia and diarrhoea patients, costs of commodities to treat the diseases in the open market, and use of treatment guidelines issued by the WHO.

2.4. Data Collection for Budgetary Gap Analysis

2.4.1. Current level of funding commitment by the government

To calculate current level of funding on procurement of commodities for treatment of diarrhoea and pneumonia, the following data was collected from provincial health

⁵ Pakistan National Health Accounts 2005-06

departments (including vertical health programmes), outsourced entities, finance departments, and planning and development departments:

- Budget vs expenditure for the past three years on all types of drugs and medicines (the source was the computerised budgeting and accounting system installed at respective Finance Departments and accounting / treasury offices throughout the country), and
- Quantities and unit costs of procurement (ORS, Zinc syrup - 60ml and 90ml, and Amoxicillin syrup - 125mg and 250mg).

Quantitative records have been obtained from either IT systems (including procurement system in place in Government of Punjab) or manual registers.

2.4.2. Determination of forecasting mechanism for ascertaining funding requirements

To forecast estimate funding requirements, the following data were used:

1. Standard treatment guidelines issued by WHO for Pakistan in 2006.
2. Total number of children under the age of 5 in Pakistan were estimated using the population census data and the age cohort data published by the Pakistan Bureau of Statistics. Two methods were used to estimate the number of under 5 children:
 - a. Age cohort data of 1998 was applied to the provisional 2017 census numbers, and
 - b. Population growth rates of 1998 were taken and population estimated for 2016. Age cohort data of 1998 was applied to estimated population of 2016.

Since provisional 2017 census numbers are being challenged by different governments, it is not clear if these numbers will remain the same in the future. Hence, above two methods have been used.

3. To estimate disease burden of diarrhoea and pneumonia in under 5 children, data from PDHS 2012-13 was used.
4. The following types of data for children under 5 were used from the above surveys:
 - a. Diarrhoea:
 - % of children with diarrhoea incidence
 - % of children with incidence of diarrhoea with blood
 - % of children sought treatment from public and private health providers
 - % of children who did not receive any treatment
 - % of children who were provided with ORS and Zinc supplements
 - % of children who were provided ORS and Zinc by public sector health providers
 - b. Pneumonia:
 - % of children with symptoms
 - % of children sought treatment from public and private health providers

- % of children who were provided with antibiotics
- % of children who were provided antibiotics by public sector health providers.

2.4.3. Method for calculating direct and indirect costs

Since the government's budgeting and accounting system does not use the system of apportionment of costs to commodities for treatment of the two diseases, the following method has been used to estimate related labour and overhead costs:

- Average pay and allowances of all government officials working in the health sector (including health departments, and health programmes) were reviewed. In addition, the number of people associated with supply chain activities (e.g. procurement planning, bidding and vendor selection, supply and distribution, storage, etc.) were estimated.
- For the overhead costs, budgets of repairs and maintenance, operating expenditure and depreciation of costs of buildings, and other assets were analysed.
- Based on the above, the following percentages have been used to estimate related labour and overhead costs:
 - Labour costs estimated at 3% of the commodity costs
 - Overhead costs estimated at 5% of commodity costs.

2.4.4. Method for estimating cost of private sector treatment (out of pocket expenditure)

To estimate out of pocket expenditure incurred on treatment of diarrhoea and pneumonia, the following method has been used:

Total quantitative requirement of morbid population X (times) percentage of out of pocket expenditure as provided in National Health Accounts of 2005-06 X (times) cost of commodities and treatment in the private sector.

2.4.5. Commodities Provided by Donors

In addition to expenditure carried out by the public sector (government, public sector enterprises, etc.), the health care services are also provided by the development partners, non-governmental organisations (NGOs), and the private sector in the country.

Since the largest amount of expenditure is carried out by the government, this study will predominately review the government's budget⁶ on procurement of drugs and supplies. In addition, efforts will also be made to gather information on procurement of drugs and supplies for diarrhoea and pneumonia by the development partners (e.g. USAID, WHO, UNICEF, etc.).

⁶ Including budget spent through vertical health programmes (e.g. Lady Health Workers), budget spent through outsourced service delivery (e.g. Rural Support Programme, People Primary Health Care Initiative, etc.), and sub-provincial levels (e.g. District Health Officers, CEOs in districts in Punjab)

2.5. Study of Budget performance and implementation challenges

To understand the budget performance and implementation challenges at provincial and sub-provincial levels - key informant interviews were held with 44 health facilities in 10 districts in Punjab and Sindh were undertaken. In addition, interviews were held with provincial finance departments (including vertical health programmes), departments of health, departments of planning and development. Discussions were also held with outsourced entities (e.g. PPHI, and IHS in Sindh).

2.6. Data Caveats

While efforts have been made to collect up-to-date and relevant data, there are however, certain caveats:

1. The PDHS, MICS and PSLM surveys were not available for the latest year (2016-17) and hence it was difficult to estimate the exact disease burden of diarrhoea and pneumonia to calculate budgetary gap. In addition, these surveys presented information of disease prevalence in the past two weeks of the survey holding. A more frequent data collection would provide relatively better information.
2. While efforts were made to collect district-wise commodities distributed, this data could not be obtained from FATA, KP and Punjab. Hence, data of total commodities procured for the whole of the province was used.
3. For FATA, data of procurements through development budget could not be obtained. Only data related to quantities and unit costs of procurement of ORS was provided.

3. Study Findings

3.1. Introduction

Pakistan has one of the highest levels of infant mortality in the World. Around half of mortalities are caused by diarrhoea, pneumonia and malaria. As per the Pakistan Health and Demographics Survey of 2012-13, around 23% of children under the age of 5 had diarrhoea two weeks before the survey, 1 in 9 children suffering from diarrhoea received no treatment at all, 61% were taken to appropriate health provider, 62% of children did not receive appropriate ORS (Oral Rehydration Solution) treatments, and only 1.5% received zinc appropriately. Similarly, 16% of children under the age of 5 had acute respiratory infection (ARI) / pneumonia two weeks before the survey, 59% suffering from pneumonia receive no treatment, and 42% with ARI received antibiotics. While Pakistan has a national immunisation programme, there is currently no vaccine⁷ administered for diarrhoea and in the country.

As per PHDS, diarrhoea remains a leading cause of childhood morbidity and mortality in developing countries. Unfortunately, despite simple treatment guidelines, 53,300 children die of diarrhoea each year, and there was an average of four to six episodes of diarrhoea per child per year. Diarrhoea is a major cause of mortality and morbidity among Pakistani children despite decades of concerted efforts and special programs⁸. Similarly, around 91,000 children die of pneumonia each year in Pakistan.

The government spent⁹ between 0.5% to 0.8% of GDP annually on health over the past 10 years. Analysed as proportion of national income (GDP) and per capita investment, this level is on the lower side when compared against other regional economies having similar models of public health care services¹⁰. In addition to inadequacy of public resources, there are questions on the equity, and efficiency on the use of public funds. Out of pocket expenditure in the country is high at 65% as per National Health Accounts 2005-06.

3.2. Estimating the current level of funding commitment by government towards purchase of commodities for treatment of Pneumonia and Diarrhoea

3.2.1. Commodities currently used for the treatment

During the visits to 44 health facilities in Sindh and Punjab, it was reported that medical professionals generally prescribe the following commodities:

For treatment of diarrhoea:

1. Low-osmolarity Oral Rehydration Salts (ORS)

⁷ As part of the Extended Programme for Immunisation, the province of Punjab is piloting Rota Virus vaccine in 6 districts in Punjab. For this purpose, the Government of Punjab is providing additional budgetary resources

⁸ Pakistan Demographic and Health Survey 2012-13

⁹ Economic Survey 2016-17, Ministry of Finance - Page 187

¹⁰ WHO benchmark is health spending of 6 percent of GDP. Pakistan's per capita health spending is US\$36.2 which is much below the WHO's benchmark of US\$86 for low-income countries. Source: Economic Survey 2016-17, Ministry of Finance - Page 187

2. Zinc Sulphate (Syrup) - 60ml and 90ml
3. In a few instances, a co-packaged ORS and Zinc (containing 3 sachets of ORS and 1 bottle of Zinc Sulphate syrup)
4. In one health DHQ Zinc dispersible tablets were procured locally from the market

For treatment of pneumonia:

1. Amoxicillin Suspension (125mg and 250mg)
2. Any related antibiotic (e.g. Augmentin) in case Amoxicillin is not available at the time of prescription
3. Oxygen in case of acute Pneumonia
4. In some District Headquarters Pulse Oximeter and Oxygen Concentrator are also found to be available.

The WHO, in standard treatment guidelines for Pakistan 2006, prescribes the following:

For treatment of diarrhoea:

1. The normal prescription is 1-3 sachets of ORS and a bottle of Zinc enough for 10 - 14 days of intake
2. For Diarrhoea with blood - dysentery - 3-5 sachets of ORS and a bottle of Zinc for more than 2 weeks of intake

For treatment of pneumonia:

1. Pneumonia - Amoxicillin anti-biotic syrup - between 125mg and 250mg depending on severity
2. Use of ARI Timer and Pulse Oximeter
3. Oxygen.

The following was observed during key informant interviews with 44 health providers:

1. Not all health providers have clear understanding of standard treatment guidelines,
2. Most of the BHUs and RHCs do not have ARI Timer or Pulse Oximeter,
3. Only PPHI managed BHUs in Sindh had stock of co-packaged ORS and Zinc, and
4. In a number of cases, the health practitioners prescribed an anti-biotic together with ORS for treatment of diarrhoea.

3.2.2. Stocks of commodities

Interviews and physical checking in 44 health facilities and 5 warehouses (Table 17 and Table 18) in Punjab and Sindh revealed that, with some exceptions, all the health facilities had ORS in stock. During the visits, it was communicated by the representatives of the health facilities that ample stocks were available for the past 2 to 3 years. There were however, certain health facilities where Zinc and Amoxicillin was found to be out of stock.

It was noted both in Sindh and Punjab that in a number of cases the lady health workers (LHWs) did not have stocks of ORS and Zinc for the past few months.

During our interactions with the officials of health departments in Balochistan, Khyber Pakhtunkhwa, and FATA, it was noted that stocks outs do occur in their health facilities of the three commodities - ORS, Zinc and Amoxicillin.

3.2.3. Current funding commitments

As per the data collected from various governments, the total government funding in 2015-16 on purchase of commodities for treatment of diarrhoea and pneumonia arrive at Rs.766 million. Including direct and indirect costs the total funding was Rs.828 million.

Table 2: Current funding commitments by the Government

Geography	Government Budget Spending 2016-17 (Excl. Labour and Overhead Costs)				Personnel Costs	Overhead Costs	Govt Spending (Incl. Labour and OH Costs)
	Zinc			Total			
	ORS	Sulphate	Amoxicillin				
	Rupees in millions						
Balochistan	2.3	24.6	12.7	39.7	1.19	1.98	42.84
FATA	0.9			0.9	0.06	0.10	1.08
Federal	1.3	0.5	3.6	5.4	0.16	0.27	5.87
KP	1.7	0.5	3.6	6.2	1.69	2.81	10.74
Punjab	7.5	55.9	239.3	262.6	3.88	23.13	299.65
Sindh	5.2	23.5	24.2	52.9	6.01	10.02	68.93
Pakistan	38.46	11.93	15.85	66.24	2.99	28.31	97.54

Source: Government Budget Spending Information provided by different government organisations

3.2.4. Existing budget forecasting method

During key informant interviews with 44 health service providers, it was noted that:

- To demand budget, the consumption basis of forecasting the requirements of commodities is used. A certain percentage is added to the previous year's consumption quantities to arrive at the next year's forecast.
- The budget demanded is sent to the provincial departments of health where it is rationalised (often downward revision is applied). In addition, the Finance Departments also apply downward revision keeping in view resource requirements.
- Generally, DHIS data for morbidity is not used for estimating budget required for the next year.

While collecting data, it was noted that data of procurements was available with different organisations. There was an absence of central register that would specify the following on period basis;

- Total quantities in stock at warehouses and with service providers,
- Total quantities distributed by warehouses to service providers,
- Total quantities procured during the year using the Central Rate Contract,
- Total quantities procured using Local Purchase head of the budget, and
- Quantities procured by autonomous hospitals, and outsourced entities.

Absence of the centralised supply chain data means that the system is prone to inefficiencies.

In our visit to Mithi in District Tharparkar of Sindh, it was noted that the supplies of ORS and Zinc were provided much more than the requirements.

3.2.5. Donor Contribution

In addition to the public sector, private sector, a number of donors - both national and international provide aid in shape of donations to the government of Pakistan.

In addition to the Government, in 2016 the UNICEF supplied the Government of Pakistan with ORS, Zinc and oxygen concentrators worth \$62,076. In addition, Micro Nutrient Initiative provided 100,000 co-packaged ORS and Zinc syrup in 2016-17 to one district in Punjab.

3.2.6. Budget Demanded Vs Budget Allocated for Public Health Providers

It is learnt from the visits to 11 districts, that the budget for procurement of drugs is generally demands 15 - 20% more than the previous year's consumption by the DDOs. However, the health departments, based on availability of the budget, and their own understanding of the consumption patterns, reduce the budget demanded. This budget amount is forwarded to the Finance Department for inclusion in the budget. Since the Finance Department does not have sectoral experience, their Section Officers (in charge of finalising budget demands) further reduce the amount by allocating a certain percentage increase (usually taking into account inflation adjustments) each year.

In some cases, e.g. Balochistan, it is observed that the Finance Department itself prepares the budget of the health department by providing a certain percentage increase each year.

3.3. Funding requirements to purchase essential commodities for treatment of Pneumonia and Diarrhoea

To estimate funding requirements to purchase essential commodities for treatment of pneumonia and diarrhoea the following method has been used:

3.3.1. Calculation of population of under 5 children in Pakistan

Provisional estimates¹¹ from the recent population census suggest that Pakistan has a total population of 207.8 million. While demographic statistics of age cohort have not yet been published, using the age cohort of 1998, it is estimated that there are 27.8 million children under the age of 5 years. Based on the above population figures and number of cases, each year the number of children with diarrhoea are estimated to be 6.3 million, and children with ARI are estimated to be 4.4 million.

Table 3: Pakistan's Population

Geography	Population (millions) 2017 Census*			2017 vs 1998 Population Growth			US Children (millions) 2017 Census			Estimated Population based on 1998 growth rates (millions)**		
	Urban	Rural	Total	Urban	Rural	Total	Urban	Rural	Total	Population 1998	Population 2016	US Children
									13.4%			13.4%
Balochistan	0.4	0.9	2.3	3.5%	3.3%	3.4%	0.5	1.2	1.7	6.6	1.3	1.5
FATA	0.1	0.9	5.0	2.7%	2.4%	2.4%	0.0	0.7	0.7	3.2	5.3	0.7
Federal	1.0	1.0	2.0	3.5%	7.0%	4.9%	0.1	0.1	0.3	0.8	0.8	0.1
KP	5.7	4.8	10.5	3.0%	2.9%	2.9%	0.8	3.3	4.1	7.7	9.7	4.0
Punjab	0.4	9.6	10.0	2.7%	1.8%	2.1%	5.4	9.3	14.7	3.6	17.3	5.7
Sindh	4.9	3.0	7.9	2.5%	2.4%	2.4%	3.3	3.1	6.4	0.4	5.4	4.4
Pakistan	5.6	32.2	207.8	2.7%	2.2%	2.4%	0.1	7.7	7.8	32.4	219.9	29.5

*Provisional Results of the Population Census 2017 (<http://www.pbs.gov.pk/node/2047>)

**Estimated based on the population growth rates of census 1998

Source: Pakistan Bureau of Statistics, National Institute of Population Studies

¹¹ <http://www.pbs.gov.pk/node/2047> - The provisional numbers are being challenged by different political parties and it is not clear whether these numbers will be adjusted in the future

However, since the provisional census numbers have been challenged by different governments in the country, this study also used an alternative method to estimate population. Based on the alternative method, the total population of children is estimated at 29.5 million. The difference between the two methods, in totality, amounts to 2.1 million or around 7%. As per the provisional numbers of population census 2017, the highest amount of population growth rates has been noted in Balochistan, Federal Government, and KP.

For the purposes of this study, census results of 2017 population figures have been used to calculate total morbid population, population using public health facilities, and total value of out of pocket expenditure on treatment of diarrhoea and pneumonia.

3.3.2. Estimation of Morbid Population - All types of Diarrhoea in under 5 children

Using the provisional numbers of population census of 2017, age cohort from 1998 population census, and the PDHS 2012-13 statistics, the total number of children with diarrhoea are estimated to be around 6.3 million in Pakistan:

Table 4: Number of Children Under 5 Years of age with All Diarrhoea (% and Number)

Disease Burden and Treatment of All Diarrhoea (under 5 children) %												
Geography	U5 with Diarrhoea			Treatment sought (public and private)			ORS provided			Zinc supplements provided		
	Urban	Rural	Total	Urban	Rural	Total	Urban	Rural	Total	Urban	Rural	Total
Balochistan	11.3%	12.2%	12.1%			43.4%			41.5%			0.8%
FATA	21.6%	29.1%	27.9%			23.0%			35.5%			2.3%
Federal			20.5%			66.5%			53.9%			2.5%
KP	21.6%	29.1%	27.9%			23.0%			35.5%			2.3%
Punjab	22.5%	21.6%	21.9%			68.6%			35.2%			1.5%
Sindh	21.9%	23.9%	23.1%			73.0%			45.2%			1.0%
Pakistan	21.9%	22.7%	22.5%	72.3%	56.4%	61.0%	41.5%	36.6%	38.0%	1.5%	1.5%	1.5%
Lowest wealth quintile			22.8%			54.0%			33.6%			2.2%
Source: PDHS 2012-13 Pakistan Bureau of Statistics												

Disease Burden and Treatment of All Diarrhoea (under 5 children) Millions of Children												
Geography	Estimated U5 Population			U5 with Diarrhoea			Treatment sought			ORS provided		
	Urban	Rural	Total	Urban	Rural	Total	Urban	Rural	Total	Urban	Rural	Total
Balochistan	110.5	110.2	110.7	110.1	110.1	110.2			110.1			110.0
FATA	110.0	110.7	110.7	110.0	110.2	110.2			110.0			110.0
Federal	110.1	110.1	110.3	110.0	110.1	110.1			110.0			110.0
KP	110.8	110.3	110.1	110.2	110.0	110.1			110.3			110.0
Punjab	110.4	110.3	110.4	110.2	110.0	110.2			110.2			110.0
Sindh	110.3	110.1	110.4	110.7	110.7	110.5			110.1			110.0
Pakistan	110.1	110.7	110.8	110.2	110.0	110.3	110.3	110.0	110.8	110.2	110.5	110.5
Lowest wealth quintile						110.3			110.4			110.1
Source: Census 2017 Provisional Data and Age Cohort from 1998 population census, PDHS 2012-13 Pakistan Bureau of Statistics												

Out of 6.3 million children with diarrhoea (22.5%) - around 3.4 million sought treatment (61%) and 1.2 million children (38%) were provided with ORS, and 0.1 million children (1.5%) were provided with Zinc supplements. As per PDHS around 4 - 6 episodes of diarrhoea are recorded each year.

MICS is currently not available

3.3.3. Estimation of Morbid Population - Bloody Diarrhoea in under 5 children

As per PDHS, children with bloody diarrhoea are around 0.7 million (2.5% of the population of children under 5).

Table 5: Number of Children Under 5 Years of age with Bloody Diarrhoea (% and Number)

Disease Burden and Treatment of Diarrhoea with blood (under 5 children) %				Disease Burden of Diarrhoea with blood (under 5 children) Millions of Children			
Geography	US with Diarrhoea with blood			Treatment sought	ORS provided	Zinc provided	
	Urban	Rural	Total				
Balochistan	0.9%	1.3%	1.2%				
FATA	1.6%	1.9%	1.9%				
Federal			0.9%				
KP	1.6%	1.9%	1.9%				
Punjab	1.8%	2.8%	2.5%				
Sindh	1.5%	3.1%	2.5%				
Pakistan	1.7%	2.6%	2.3%	71.1%	50.2%	1.0%	
Lowest wealth quintile			2.5%	54.0%	33.6%	2.2%	

Source: PDHS 2012-13 Pakistan Bureau of Statistics

3.3.4. Estimation of Morbid Population - Acute Respiratory Infection (ARI) / Pneumonia in under 5 children

Using the provisional numbers of population census of 2017, age cohort from 1998 population census, and the PDHS 2012-13 statistics, the total number of children with ARI / pneumonia are estimated to be around 4.4 million in Pakistan:

Table 6: Number of Children Under 5 Years of age with ARI (% and Number)

Disease Burden and Treatment of Pneumonia Symptoms %									
Geography	Pneumonia Symptoms			Treatment sought			Anti-biotic received		
	Urban	Rural	Total	Urban	Rural	Total	Urban	Rural	Total
Balochistan	12.1%	9.1%	9.7%	77.8%	46.3%	53.5%	38.9%	18.2%	23.0%
FATA	24.9%	23.1%	23.4%	45.5%	25.9%	29.3%	43.9%	46.3%	45.9%
Federal			8.9%			66.9%			32.8%
KP	24.9%	23.1%	23.4%	45.5%	25.9%	29.3%	43.9%	46.3%	45.9%
Punjab	13.5%	16.1%	15.8%	79.3%	69.7%	72.1%	43.9%	44.5%	44.3%
Sindh	14.5%	11.6%	12.8%	80.4%	82.6%	81.6%	28.2%	34.4%	31.6%
Pakistan	16.4%	14.6%	15.9%	75.1%	60.4%	64.4%	38.9%	42.5%	41.5%
Lowest wealth quintile			13.4%			56.6%			41.8%

Source: Pakistani Bureau of Statistics

Disease Burden of Pneumonia Symptoms Millions of Children												
Geography	Estimated US Population			Pneumonia Symptoms			Treatment sought			Anti-biotic received		
	Urban	Rural	Total	Urban	Rural	Total	Urban	Rural	Total	Urban	Rural	Total
Balochistan	0.5	0.2	0.7	0.1	0.1	0.2	0.0	0.1	0.1	0.0	0.0	0.0
FATA	0.0	0.7	0.7	0.0	0.2	0.2	0.0	0.0	0.0	0.0	0.1	0.1
Federal	0.1	0.1	0.3									
KP	0.8	0.3	1.1	0.2	0.8	1.0	0.1	0.2	0.3	0.1	0.4	0.4
Punjab	0.4	0.3	0.7	0.7	0.5	1.3	0.6	0.0	0.7	0.3	0.7	1.0
Sindh	0.3	0.1	0.4	0.5	0.4	0.8	0.4	0.3	0.7	0.1	0.1	0.3
Pakistan	0.1	0.7	0.8	0.7	0.6	1.4	0.2	0.6	0.9	0.6	0.1	0.8
Lowest wealth quintile						0.7			0.1			0.6

Out of a total of around 3.7 million children with pneumonia, around 2.1 million seek treatment and around 1.6 million are provided with an antibiotic.

3.3.5. Number of diarrhoea and pneumonia cases in a year

Based on 4 episodes of diarrhoea¹², there are on average 25 million cases of all types of diarrhoea, and around 2.5 million cases of bloody diarrhoea in the country. Based on 6

¹² Source: PDHS 2012-13 - on average 4 - 6 episodes of diarrhoea are reported each year

episodes of diarrhoea, there are on average 38 million cases of all types of diarrhoea, and around 3.8 million cases of bloody diarrhoea in the country.

Similarly, based on 1 episode of pneumonia there are 4.4 million cases and around 9 million cases in case of 2 episodes in the country.

For lowest income quantile, the number of diarrhoea cases can be as high as 38 million and around 4.2 million for bloody diarrhoea. Number of cases for lowest income quantile can be as high as 7.5 million cases in case of two episodes of diarrhoea.

The total number of children with diarrhoea are around 6.3 million and 4.4 million with pneumonia.

Table 7: Number of Morbid Cases - Diarrhoea and Pneumonia

Under-5 Mortality and Morbidity												
Geography	U5 Mortality Per 1,000 live births	Data Year	U5 Children estimated in 2017	U5 Children with all types of diarrhoea	U5 Children with bloody diarrhoea	U5 Children with Pneumonia symptoms	Cases in U5 Children					
							4 episodes per yr	6 episodes per yr	4 episodes per yr	6 episodes per yr	1 episode per yr	2 episodes per yr
Balochistan	0.58	2003-04	0.7	0.20	0.02	0.16	0.80	1.20	0.08	0.12	0.16	0.32
FATA	0.04	2007-08	0.7	0.19	0.01	0.16	0.75	1.12	0.05	0.08	0.16	0.31
Federal			0.3	0.06	0.00	0.02	0.22	0.33	0.01	0.01	0.02	0.05
KP	0.75	2006-07	1	0.14	0.08	0.96	0.56	0.85	0.31	0.47	0.96	1.91
Punjab	0.04	2010-11	4.7	0.23	0.37	0.33	2.91	9.37	0.47	0.21	0.33	0.66
Sindh	0.12	2003-04	6.4	0.48	0.16	0.82	0.93	0.89	0.64	0.96	0.82	1.64
Pakistan	0.89	2012-13	27.8	5.26	0.64	4.43	25.06	37.59	2.56	3.84	4.43	8.85
Lowest wealth quantile				6.35	0.70	0.73	5.4	8.1	2.78	4.18	3.73	7.46

Source: PDHS 2012-13 and Pakistan Bureau of Statistics

3.3.6. Morbid population in under 5 years of children using public health facilities

Data of morbid population (in under 5 children) using public health facilities is not available in PDHS 2012-13. The only reliable estimate is available from PLSM 2014-15. The percentage of population using public health facilities is applied to the population census data of 2017.

As per PLSM 2014-15, the morbid population (children under the age of 5 with diarrhoea and pneumonia) using public health facilities was around 31%.

This means that out of a total of 6.3 million children with diarrhoea, around 2 million visited public health facilities out of which 0.7 million received ORS, 0.03 million children received zinc sulphate syrup. For pneumonia, out of 4.4 million children with pneumonia symptoms around 1.1 million visited public health facilities and 0.6 million children received an anti-biotic.

Table 8: Morbid Population Using Public Health Facilities

Morbid population using public sector health facilities											
Geography	U5 Children with all types of diarrhoea	U5 Children visited public health facility	U5 Children received treatment	ORS provided	Zinc Sulphate provided	U5 Children with Pneumonia symptoms	U5 Children visited public health facility	Anti-biotic provided			
	millions	%	millions	%	millions	millions	millions	millions	%	millions	millions
Balochistan	0.20	48.8%	0.10	82.4%	0.08	0.07	0.00	0.16	48.8%	0.08	0.02
FATA	0.19	50.6%	0.09	85.2%	0.08	0.03	0.00	0.16	50.6%	0.08	0.04
Federal	0.06	24.0%	0.01	90.7%	0.01	0.01	0.00	0.02	24.0%	0.01	0.00
KP	0.14	50.6%	0.58	85.2%	0.49	0.17	0.01	0.96	50.6%	0.48	0.22
Punjab	0.23	20.5%	0.66	89.6%	0.59	0.21	0.01	0.33	20.5%	0.48	0.21
Sindh	0.48	32.3%	0.48	92.8%	0.44	0.20	0.00	0.82	32.3%	0.26	0.08
Pakistan	5.26	30.7%	1.92	88.5%	1.70	0.65	0.03	4.43	30.7%	1.36	0.56
Lowest wealth quantile	6.35	30.7%	1.95		1.72	0.58	0.04	0.73	30.7%	1.1	0.48

Source: PDHS 2012-13 and PLSM 2014-15 and Pakistan Bureau of Statistics

Based on morbid population numbers and the number of episodes of the two diseases (using PDHS 212-13), the total quantitative requirement of the three commodities (ORS, Zinc Sulphate syrup and Amoxicillin suspension) in a year is calculated as:

- Table 9: Quantitative Requirement for Treatment of Diarrhoea and Pneumonia in Pakistan

Source: PDHS 2012-13 and provisional number of census 2017, Pakistan Bureau of Statistics

Since the government buys in bulk, it is able to gain benefits due to economies of scale. Average prices of commodities are as follows:

Prices of Commodities	ORS (Sachet)	Zinc Sulphate (Syrup) 90ML	Amoxicillin (Syrup) 125-250mg
	Rs. Per sachet	Rs. Per bottle	Rs. Per bottle
Average Price of Government Procurement	6	25	45
Available in the Market:			
Price Range	₹18	₹180	₹250
Average	10	80	150
Source: Tender Contracts, Procurement Records of Government			
Source: Pharmaceutical Outlet			

3.4. Budgetary Gap

Assuming that the government picks up the cost of treatment of all diarrhoea and pneumonia morbid population under the age of 5, the total cost requirement arrives at Rs.2.2 billion per year.

The assumptions used in the tables below are based on data collected and analysed in the previous section.

Table 11: Budget Required - Assuming the Government provides Treatment to all Diarrhoea and Pneumonia patients under the age of 5 years - Rs. Millions

Requirement of Budget for the Treatment of Diarrhoea and Pneumonia									
Geography	ORS Sachet	Zinc Sulphate (Syrup) 50-90ML	Amoxicillin (Syrup) 125-250mg	Diarrhoea Treatment				Pneumonia Treatment	
				ORS		Zinc Sulphate (Syrup)		Amoxicillin (Syrup)	
				Sachets		1 Bottle		1 Bottle	
				4 episodes per yr	6 episodes per yr	4 episodes per yr	6 episodes per yr	1 episode per yr	2 episodes per yr
				Average cost per Unit in Pak Rupees				Millions of Rupees	
Balochistan	6	25	45	19.2	28.8	0.0	54.0	7.2	14.4
FATA	6	25	45	8.0	6.9	8.7	60.5	7.1	4.1
Federal	6	25	45	5.3	9.9	5.5	4.9	1.1	2.2
KP	6	25	45	09.6	64.3	14.1	08.1	3.1	6.1
Punjab	6	25	45	09.9	64.9	22.8	71.7	04.8	09.6
Sindh	6	25	45	42.3	13.4	48.2	00.2	7.0	3.9
Pakistan	6	25	45	601.4	502.1	26.4	691.4	99.2	98.4
Lowest wealth quantile				609.4	514.1	34.8	52.2	67.9	35.8
Source: PDHS 2012-13, Population Census 2017, Average Prices from Central State Contracts, Procurements									

Source: PDHS 2012-13, Population Census 2017, Average Prices from Central Rate Contracts and Procurements

Table 12: Budgetary Gap - Cost of Medicines - Rs. Millions

Budgetary Gap in Medicine Cost									
Geography	Diarrhoea Treatment			Pneumonia Treatment	Government Budget Spending 2016-17				
	Zinc			Total	Zinc Amoxicillin				Gap
	ORS	Sulphate	Amoxicillin		ORS	Sulphate	in	Total	
	Millions of Rupees								
Balochistan	4.0	7.0	0.8	1.9	2.3	4.6	2.7	9.7	32.2
FATA	2.4	4.6	0.6	7.6	9			9	65.7
Federal	5.6	0.2	6	8.4	3	0.5	6	5.4	3.0
KP	36.9	11.1	4.6	12.7	12.7	7.5	6.0	56.2	356.4
Punjab	87.4	97.3	57.2	141.9	67.5	55.9	239.3	462.6	679.2
Sindh	77.9	74.2	55.4	507.5	52.7	3.5	24.2	200.3	307.2
Pakistan	51.7	158.9	98.8	2,209.5	38.5	11.9	15.8	66.2	443.2
Lowest wealth quantile	61.8	93.5	51.8	807.1					1,040.8

Source: Government Budget Spending Information provided by different Government Organisations

The government is currently spending Rs.766 million on the commodities for the treatment of diarrhoea and pneumonia per year. Adding the labour and overhead costs the total requirement is estimated at around Rs.2.3 billion:

Table 13: Cost requirements including Labour and Overhead Costs - Rs. Millions

Geography	Medicine cost	Estimated Labour Cost	Estimated Overhead Cost	Required cost
		% of medicine cost	% of medicine cost	
	Rs. Million	cost	cost	Rs. Million
Balochistan	1.9	3.0%	5.0%	4.8
FATA	7.6	3.0%	5.0%	70.3
Federal	8.4	3.0%	5.0%	9.2
KP	12.7	3.0%	5.0%	29.2
Punjab	141.9	3.0%	5.0%	187.6
Sindh	507.5	3.0%	5.0%	527.8
Pakistan	2,209.5	3.0%	5.0%	2,297.8
Lowest wealth quantile	807.1	3.0%	5.0%	879.4

Based on these calculations the total budgetary gap, if the entire morbid population is treated by the government, comes out to be around Rs.1.5 billion.

3.5. Out of Pocket Expenditure

As per the National Health Accounts of 2005-06 on average around 65 percent of expenditure is incurred out of pocket in Pakistan.

The out of pocket expenditure is expensive as it attracts three types of costs:

1. Cost of inspection,
2. Cost of treatment, and
3. Cost of travel.

Out of the above three, the cost of treatment is expensive as the average price of commodities for the treatment of diarrhoea and pneumonia is higher (Ref. Section 3.3.8) in the market as compared to that incurred by the government.

Based on PDHS survey, and National Health Accounts, population census of 2017, and average prices of commodities in the market, the out of pocket expenditure on the treatment of diarrhoea and pneumonia is estimated to be Rs.3.1 billion in a year.

Table 14: Out of Pocket - Cost of Treatment in Under 5 children - Rs. Millions

Cost@PrivateSector	Treatment@DiarrhoeaAndPneumonia@Under5@children												Out@of@pocket@expenditure@on@health@%	Treatment@DiarrhoeaPneumonia								Total@average@costs@Millions						
	Diarrhoea@treatment				Pneumonia@treatment				Diarrhoea@treatment					Pneumonia@treatment														
	ORS		ZincSulphate@Syrup		Amoxicillin@Syrup		ORS		ZincSulphate@Syrup		Amoxicillin@Syrup			ORS		ZincSulphate@Syrup		Amoxicillin@Syrup										
	3-5@Sachets		1@Bottle		1@Bottle		3-5@Sachets		1@Bottle		1@Bottle			3-5@Sachets		1@Bottle		1@Bottle										
	4@episodes@per@yr	6@episodes@per@yr	4@episodes@per@yr	6@episodes@per@yr	1@episode@per@yr	2@episodes@per@yr	4@episodes@per@yr	6@episodes@per@yr	4@episodes@per@yr	6@episodes@per@yr	1@episode@per@yr	2@episodes@per@yr		4@episodes@per@yr	6@episodes@per@yr	1@episode@per@yr	2@episodes@per@yr	4@episodes@per@yr	6@episodes@per@yr									
	millions													Cost@RS@Millions														
Balochistan	0.20	0.80	0.75	0.20	0.16	0.32	29.1	9.3	4.0	8.7	28.0	7.0	4.0	25.5	2.99	4.49	0.80	1.12	0.16	0.31	4.7	5.3	8.0	0.7	6.0	9.9	9.8	24.9
FATA	0.88	3.32	0.22	0.33	0.02	0.05	5.3	5.8	6.6	1.5	7.3	2.3	7.7	25.1	0.88	3.32	0.22	0.33	0.02	0.05	5.3	5.8	6.6	1.5	7.3	2.3	7.7	25.1
KP	8.26	7.39	4.56	6.85	0.96	1.91	4.7	54.6	23.9	309.2	463.8	21.6	43.1	62.1	8.26	7.39	4.56	6.85	0.96	1.91	4.7	54.6	23.9	309.2	463.8	21.6	43.1	62.1
Punjab	1.65	7.48	2.91	9.37	0.33	0.66	59.1	57.1	35.7	14.3	071.4	241.6	483.1	701.6	1.65	7.48	2.91	9.37	0.33	0.66	59.1	57.1	35.7	14.3	071.4	241.6	483.1	701.6
Sindh	3.72	5.57	5.93	8.89	0.82	1.64	5.27	24.9	87.3	249.8	374.7	64.9	29.8	565.7	3.72	5.57	5.93	8.89	0.82	1.64	5.27	24.9	87.3	249.8	374.7	64.9	29.8	565.7
Pakistan	0.23	50.35	25.06	37.59	4.43	8.85	5.3	54.9	98.24	309.8	964.7	433.9	867.7	106.7	0.23	50.35	25.06	37.59	4.43	8.85	5.3	54.9	98.24	309.8	964.7	433.9	867.7	106.7
Lowest@quantile	01.57	52.35	25.39	88.09	7.73	14.46									01.57	52.35	25.39	88.09	7.73	14.46								

Source: PDHS 2012-13, Provisional Population Census 2017, National Health Accounts 2005-06

Source: PDHS 2012-13, Provisional Population Census 2017, National Health Accounts 2005-06

There is a need to reduce this level of expenditure especially in the lowest quantile of the population in under 5 years of children.

3.5.9. Role of the private sector

Role of the private sector in health service delivery is significantly large in Pakistan.

Discussions with 5 private sector general physicians revealed the following:

- There is generally lack of awareness of standard treatment guidelines especially use of Zinc sulphate and is often not prescribed
- Similarly, there is generally lack of awareness of Amoxicillin being the preferred medicine for the treatment of pneumonia. Use of different anti-biotics for treatment of pneumonia is in vogue.

3.6. Budget performance and implementation challenges

During key informant interviews with provincial authorities, the following budget performance and implementation challenges were noted across the board:

- Budget demanded by the health service providers is higher than what is normally awarded to them through the budget. A two-staged rationalisation exercise takes place; one at the level of the central departments of health, and second at the level of finance departments.
- Delays occur at the time of budget releases. A number of procurement officials reported to have received substantial portion of budget releases after quarter 2 of a fiscal year. This results in delays in procurements. In some cases, it was noticed that delayed releases resulted in surrender (non-utilisation) of budgeted amounts. It was also reported in some cases that manufacturers refused to supply commodities against work orders released in late May or early June.
- Budget for procurement of medicines is funded through recurrent and development budgets. Budgets of DDOs are transparent and the heads of 'procurement of drugs and medicines' and 'procurement of equipment' are clearly shown. However, information of procurement of the type of drugs and medicines within the budget is not provided. Budget provided to outsourced entities (e.g. PPHI, IHS in Sindh) are provided in a single-line. There is an issue of transparency as it is not clearly how much funds are to be allocated by these outsourced entities on procurement of medicines. Similarly, on the development side, projects also normally include single-line budgets.
- Information on amount that has been allocated and approved as part of the budget is provided roughly 3-4 months after the approval by the Provincial Assembly to the spending units. Secondly, the funds are often reported to have released late resulting in delays in issuance of procurement work orders and subsequent delivery.
- Procurement officials have limited knowledge of alternative forecasting methods that can be used to forecast medicine procurements.
- In 2010 the Council of Common Interest agreed that the Federal Government would provide funding for vertical health programmes till the next National Finance Commission Award. However, in 2017 the Federal Cabinet has decided to gradually reduce funding for vertical health programmes. Going forward, this will raise challenge for the provinces to allocate sufficient funding from their budget.

In addition to the above, the following challenges were noted during key informant interviews:

3.6.1. Sindh Province

1. Recently the Sindh Government has changed the system of allocation of budgets for the 'procurement of drugs and medicines'. Until 2015-16 the budget under this head was allocated to DDOs and hence it was possible to ascertain how much budget was allocated to each district. From 2016-17 the budget is being allocated to divisional heads. This has reduced visibility and transparency of the budget allocated at the district and sub-district levels.
2. In 2015, a process of district action plans was started in the district government with the help of USAID. The district action plans, prepared with the help of consulting support, formed the basis of allocation of scarce budgetary resources as per the priorities of the districts. It was a system through which incrementalism in the budget was replaced with needs and priority based allocations. This system however, was later discontinued with

the completion of the project. The government currently does not have the capability to implement the system of formulation of district action plans on its own.

3. There were budget releases issues encountered by IHS - the organisation that manages Rural Health Centres in Sindh. For this reason, the RHCs visited that were managed by IHS did not have required stocks of medicines available.

3.6.2. Punjab Province

1. The Punjab Government has recently established a Health Facility Management Company to oversee the management of Basic Health Units. Earlier BHUs in a large number of districts were managed by the PRSP (Punjab Rural Support Programme). The PRSP had been operational for the past few years and their budgetary management systems have matured over the years. The creation of a new entity is likely to see disruptions in management of the budget, procurements, and maintenance, due to its recent establishment.
2. Lately, the Punjab Government has established District Health Authorities, that are headed by Chief Executive Officers (CEOs). The budget for procurement is reportedly being provided to these newly created authorities. Separate bank accounts are made operational where initial funds have been transferred by the provincial Finance Department. It has been informed that the provincial government is struggling to appoint relevant personnel in these authorities. Till the full appointment of required staff members, the management of the budget, the procurements, are required reporting is likely to suffer.

3.6.3. Balochistan Province

1. While the budget is allocated to DDOs in the districts, the actual procurement is undertaken by the Medical Stores Depot (MSD). This means that the funds are re-appropriated (the powers of re-appropriation of funds from one entity to another lies with the Finance Department) from all the DDOs to the MSD during the year. Only after this process and the process of funds release, can the MSD undertake procurements. It is reported that normally the first procurement order is made in February / March each year. Since the funds have to be used by 30th June each year, this leaves less time to finalise all processes.
2. The Finance Department compiles the budget (on the recurrent side) on behalf of health department. This means that what is actually required at the district level, their priorities, and the changes to incremental budget, are not accommodated.

3.6.4. Khyber Pakhtunkhwa Province

Information of quantities of procurement are not available with the health department at the time of budget formulation. Hence, largely the incremental budget mechanism is applied to formulate central rate contract.

3.6.5. FATA Secretariat

Federally Administered Tribal Areas (FATA) Secretariat undertakes procurement of medicines through the non-development as well as development budget. There is a project (of Annual Development Programme) through which amounts are spent to procure medicines in the Agencies and Frontier Regions. The funds releases through development budgets are dependent on releases made by the Planning Commission at the Federal level

and sub-divided into projects by FATA Secretariat. There are therefore, high chances of delays in budgetary funds releases and procurements.

3.7. Use of budget and the system of procurement

During key informant interviews, it was noted that different types of procurement models are in place in different provinces. The following table summarises the information obtained:

Table 15: Procurement budget at different tiers of the Government

Federal / Provincial	Procurement Authority	Non-Development budget for Procurement of Medicines at primary and secondary level
Sindh	Decentral (rates fixed through Central Rate Contract)	From 2016-17 onwards, the budget is made available at the Divisional level
	Entity level	Where provision of health services is outsourced, each individual entity is authorised to undertake procurements. These entities include; PPHI (President's Primary Health Care Initiative) - managing the BHUs in Sindh, IHS (Integrated Health Services) - managing certain RHCs in Sindh
	Combat Diarrhoea Disease programme	Budget is provided through development budget and the project director is authorised to undertake procurements
	Public private partnerships	Public private partnerships are contracts entered by the government with private sector health providers - specific examples include PPHI - President Primary Health Care Initiative managing basic health units, and HIS - Integrated Health Services managing a number of rural health centres. In addition, some Tehsil Headquarter Hospitals are contracted out. In addition, in some districts various level of health facilities have also been contracted out to other private entities like HANDS and Indus Hospital.
Balochistan	Central (rates fixed through Central Rate Contract)	Budget is made available at the level of District Health Officers - however, Balochistan follows a central procurement method and hence, during the year, the budget is re-appropriated to the Medical Stores Depot
Punjab	Decentral (rates fixed through Central Rate Contract)	From 2016-17, the budget is being made available to the District Health Authorities - who are run by a Chief Executive Officer (CEO). In addition, Punjab Health Facilities Management Company (PHFMC) has been formed which will oversee the management BHUs and RHCs in the province. Budget will be allocated to PHFMC as a single-liner
Khyber Pakhunkhwa (KP)	Decentral (rates fixed through Central Rate Contract)	Budget is made available to DHOs at the district level
Federally Administered Tribal Areas (FATA)	Decentral (rates fixed through Central Rate Contract)	Budget is made available to the DHOs at the Agency and Frontier Regions level.
	Procurement Authority	Development Budget for Procurement of Medicines
Provinces, Federal, and FATA	Vertical Programmes	The heads of vertical health programmes are authorised to undertake procurement. More specifically, for diarrhoea and pneumonia - procurements are undertaken by Lady Health Workers Programme, Maternal Neo-natal and Child Healthcare programme, and Nutrition Programme
FATA	Central	Procurement through a project is undertaken at the level of DG Health Services

3.8. Summary Findings

The above findings are summarised as follows:

- Under 5 population between 27.8 (2017 census data) and 29.5 (1998 population growth formula) million.
- Morbid population in 2017 based on percentages in PDHS 2012-13 survey:
 - Children with all types of diarrhoea are estimated at 6.3 million
 - Children with pneumonia symptoms are estimated at 4.4 million
- As per the existing medicines procured, the number of disease episodes, and rates at which the government procures commodities:
 - The total budgetary requirement to treat morbid population in Pakistan is calculated as Rs.2.2 billion in 2016-17
 - Quantity wise this translates into:
 - 125 million sachets of ORS
 - 31.5 million bottles of Zinc syrup (60ml and 90ml)
 - 6.6 million bottles of Amoxicillin suspension (125mg/5ml and 250mg/5ml)
- The Government (including outsourced entities) – spent Rs.766 million in 2016-17 on procurement of these commodities.
- Based on the above calculation – government procured medicines that catered for 35% of the total requirement.
- Government uses consumption based method of forecasting of medicines. Actual procurement is undertaken as per budget availability.
- Visits to 44 health facilities revealed that Drawing and Disbursing Officers (DDOs) – generally receive 15% - 20% less budget than demanded.
- In 10 districts, stocks of ORS, Zinc Sulphate (syrup) and Amoxicillin (suspension) in warehouses and health service providers were satisfactory, except:
 - RHCs run by IHS in Sindh – we were told by IHS that required budget was not released in 2016-17
 - LHWs - both in Punjab and Sindh – a number of LHWs reported having zero stock
 - In some cases – stock was not available due to delays on behalf of manufacturers – e.g. THQ Rohri, THQ Kandh
- Provincial authorities of Balochistan, KP and FATA informed that stock-outs occur for commodities required for treatment of diarrhoea and pneumonia in districts / agencies.
- The Federal Cabinet has approved the proposal to gradually reduce funding for provincial vertical health programmes – this raises a challenge for provinces.
- Delayed releases of funds by the Finance Departments to DDOs – result in delayed procurements – in some cases manufacturers refused to supply due to delayed work orders – resulting in budget lapse.
- Finalisation of Central Rate Contracts took place in quarter 2 of 2016.
- In Balochistan, FD prepares budget on behalf of Health DDOs.

4. Recommendations and Action Plan

Recommendations

Based on the study finding, the following key recommendations are made:

4.1.1. Improvement in forecasting mechanism

At present, the forecasting of medicines procurement is based on method of consumption. This means that a certain percentage is added to prior year's amounts consumed. The government should build its capacity to gradually move towards WHO recommended forecasting methods. Three methods can be considered by the government:

- Stock adjustment method - in this method future procurement take into account the situation of functional stock-outs,
- Morbidity based forecast - this method uses data from surveys to calculate morbid population, percentage of population using health facilities, and forecast quantities required, and
- Patient load analysis - based on DHIS data, the forecasts of quantities required are undertaken for public health facilities.

However, implementation of the above methods will require capacity enhancement of procurement entities, the departments of health, and departments of finance.

In addition, the government should consider formalising the process of development of 'district action plans' as basis of budget allocation. Under this method, each year, the district officials take stock of the following:

- Status of health facilities,
- Current level of stocks of drugs and medicines,
- Disease burden based on a representative sample survey,
- Understanding of out of pocket expenditure,
- Storage capacities,
- Understanding of health service providers - public and private, and
- Budgets made available by the government.

4.1.2. Improving efficiency in the use of government budgets

The gap for treatment between what is currently available and what is required is around 3 times. This means that the government will have to spend 3 times more budget than it is spending currently in order to treat all morbid population under the age of 5.

However, the government can reduce this ratio through a number of mechanisms:

- **Increasing efficiency in public health expenditure** - more specifically, the government can increase efficiency in expenditure if:
 - The use of co-packing of ORS and Zinc is encouraged

- The government moves towards the use of dispersible tablets of Zinc and Amoxicillin. Through the use of tablets, the average overhead costs - e.g. transportation, and warehousing - are likely to reduce
- Use of ARI Timers and Pulse Oximeters are likely to enhance patient diagnosis.
- **Government invests funds in prevention of diarrhoea and pneumonia through vaccines such as PCV and ROTA virus** - however, it is beyond the scope of this study to study the costs and efficiency levels of vaccines.
- **Governments investments in education of mothers, water and sanitation to prevent diarrhoea etc.**

4.1.3. An early completion of Central Rate Contracts

During the study visits to 11 districts, it was noted that rate contracts were generally finalised in the months of November and December 2016. For this reason, in many districts the procurement process started late. In some cases, the procurement order was forwarded to the medicine manufacturing companies in the month of May 2017. In such cases, the manufacturing companies refused to supply required medicines by June 2017.

For this reason, it is recommended that the government should aim to complete its central rate contract process ideally by 31st of July and maximum by 31st August of each year. To complete this process, efforts on quantitative procurement requirements, and finalisation of bidding documents will need to be finalised by first week of July each year.

Information of the central rate contracts should be communicated to procurement entities (DDOs) before the end of the first quarter (September) each fiscal year.

4.1.4. Procurement and distribution system in case of vertical health programmes

During the study visits to 10 districts, it was noted that in a number of cases, the lady health workers did not have the required supply of ORS, Zinc sulphate syrup and Amoxicillin suspension. In addition, since the Federal Government is likely to withdraw from its obligation to adequately fund the vertical health programmes, the responsibility would now fall on the provincial governments to appropriately fund these programmes.

The provincial governments have moved towards integrating lady health workers' programme, community mid-wives' programme and nutrition programme. Where this transition is yet to complete in all provinces, it is recommended that steps should be taken to integrate these programmes.

Three key recommendations are made for vertical health programmes:

1. The provincial governments should review the standard supply of stocks to each lady health worker. The system of procurement - from planning to end usage needs to be reviewed so that sufficient stock availability can be ensured in the communities especially in rural areas. The method of review can include:
 - a. Situation of stock-outs,
 - b. Disease burden in districts especially in rural districts (Refer to annexure on district data), and
 - c. Seasonal pattern of diseases.

2. An analysis of budget required by the vertical health programmes and how will the provincial governments take over this responsibility needs to be undertaken.
3. An IT system should be developed through which stock levels with Lady Health Supervisors can be monitored on regular basis.

4.1.5. An increase of budget on procurement of medicines for treatment of diarrhoea and pneumonia

During the study visits to 10 districts, it was noted that the DDOs often request 15 - 20% more budget than what was allocated. These budget requirements are rationalised by the health departments and later by the Finance Departments in the provinces.

Health service providers in the 10 districts noted that increase in supply of medicines, and provision of qualified doctor often leads to improved visiting patients.

For this reason, the health departments and finance departments should be consulted and requested to provide an increase of 10 - 15% in the budget head 'A03927' - procurement of drugs and medicines - more specifically for the procurement of ORS, Zinc syrup, and Amoxicillin syrup. The basis of increase of 10 - 15% is the additional budget that is demanded by the DDOs each year.

In addition, a policy note should be circulated from the Health Department to specify that this increase would be awarded to improve supplies of commodities for treatment of diarrhoea and pneumonia in specified health facilities. For this reason, a detailed study should be undertaken on the districts that have high disease prevalence and incidence and the stock availability of the essential commodities for the treatment of the two diseases.

4.1.6. Improvement in budget management practices

Three types of improvements are recommended in the budgetary practices:

1. Each DDO should be communicated with its budget, after approval from the legislative assembly, maximum by 15th August each year. At present a number of DDOs receive this information by October / November.
2. The Finance Departments in the provinces should be requested to amend the funds release policy and allow 100% release of funds on the 1st of July each year for the budget head A03927 titled 'procurement of drugs and medicines'.

4.1.7. Development of IT systems

Punjab government has developed a number of IT applications that is aiding it in procurements, stock monitoring, and provision and use of medicines at health facilities level.

A seminar should be arranged where Punjab government may be requested to show-case the IT systems to other provinces. A feasibility analysis should be undertaken on adoption of these IT systems in other provinces.

4.1.8. Determination of policy on reduction of out-of-pocket expenditure

The Government should formulate a policy regarding reduction of out-of-pocket expenditure. If the policy is to provide universal healthcare services to all then it should aim to plan for increase of budgetary resources together with supply chain management processes over a specified period of time.

Reduction of out-of-pocket expenditure may require the following types of strategies to be put in place to transition towards universal healthcare coverage:

1. Provision of a more holistic community based health workers,
2. A policy of increase in pre-payment mechanism - this can be done through introduction of health insurance cards for the lowest quantile including their family members. Coverage can also be extended to lower-middle-income quantile by introducing voluntary health insurance,
3. Increase in provincial level taxation in order to allow additional fiscal space for the provision of health care services,
4. Increase in budgetary allocations proportionate to allocations in other sectors,
5. Investments in patient referral system so as to avoid patient congestions at the secondary and tertiary health care facilities.

4.1.9. Holding of pre-budget consultative sessions / seminars with different stakeholders

Pre-budget discussions should be held with different policy stakeholders - including government, donors, parliamentarians, and related reform projects starting from December 2017. While the main theme of pre-budget discussions would seek to influence budget making for the year 2018-19, different types of topics could be discussed with different audience.

1. Government sector - Departments of Health and District Health Officials

- a. Review of the system of supply chain of medicines and equipment including system of forecasting of medicines procurement
- b. Improvements in budgetary releases to district health organisations, and DDOs
- c. Methods to ensure timely submission of procurement work orders by district DDOs and timely supply of medicines by manufacturers
- d. Discussions on introduction of co-packing and dispersal tablets for zinc sulphate and Amoxicillin
- e. Discussions on the policy to reduce out of pocket expenditure.

2. Parliamentarians - more specifically the local MPAs and MNAs

- a. Key statistics related to health sector in the district / constituency - including state of health facilities, availability of medic and para-medic staff, availability of medicines and equipment, and disease burden related to diarrhoea and pneumonia
- b. Measures to enhance health service delivery - and policy to treat diarrhoea and pneumonia cases
- c. Budget provided to and spent by health facilities and providers in the district / constituency.

3. Donor coordination group on reduction of diarrhoea and pneumonia

A donor coordination group can be formed to synergise efforts on reduction of diarrhoea and pneumonia. The coordination group, on regular basis, can discuss the following:

- a. Key initiatives in reduction of mortality and morbidity in under 5 children causes by diarrhoea and pneumonia
- b. Policy changes implemented and their impacts on mortality and morbidity
- c. Evaluations,
- d. Aid.

4.1.10. Promotion of the use of co-packaged ORS and Zinc Sulphate for the treatment of diarrhoea and Amoxicillin suspension for the treatment of pneumonia amongst private sector general practitioners

Role of the private sector is quite sufficient in Pakistan. The government should therefore, promote the use of standard treatment guidelines, as recommended by WHO, amongst private sector general physicians for the treatment of diarrhoea and pneumonia.

4.2. Formulation of a comprehensive policy incorporating prevention, treatment and behaviour change elements

While budgetary gap analysis is an important factor to consider for improving service delivery, a more holistic policy definition is suggested to enhance efficiency and effectiveness in the use of public money. Reduction of under 5 mortality and morbidity requires a policy framework taking into account the prevention, treatment and behavioural change aspects. This will ensure a multi-facet approach involving different sectors (e.g. education, hygiene, water and sanitation, health, economic prosperity, etc.) that use public funds.

4.3. Action Plans and Specific Tasks

Based on the report findings, and recommendations, the following action plan is recommended:

Table 16: Action Plan

Activities in the Action Plan		Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
1. Improvement in the method of Forecasting Procurement of Medicines											
1.1	Trainings on medicines forecasting methods										
1.2	Trainings on determination of district action plans and linkage with budgets										
2. An early completion of Central Rate Contracts											
3. Procurement and distribution system in case of vertical health programmes											
3.1	Review of standard supply										
3.2	Review of procurement and distribution system including timelines required										
4. An increase of budget on procurement of medicines for treatment of diarrhoea and pneumonia											
4.1	Discussions with health departments and finance departments on increase in budget for commodities for treatment of diarrhoea and pneumonia										
4.2	Discussions on co-packaged (ORS and Zinc) procurements										
5. Improvement in budget management practices											
6. Development of IT systems											
6.1	Presentation of IT systems of Punjab to other provincial governments										
6.2	Discussions with provinces on adoption of similar systems										
7. Determination of policy on reduction of out-of-pocket expenditure											
7.1	Determination of policy on universal health coverage										
8. Pre-budget consultative sessions on diarrhoea and pneumonia commodities											
8.1	With provincial departments of health and district health officials										
8.2	With provincial finance departments										
8.3	With local parliamentarians										

Table 17: Sindh Province - Availability of Stocks in Districts visited during the Field Study - June 2017

District	Health Facility	ORS	Amoxicillin syrup	Zinc Syrup
Sukkur / Sukkur	EDOH stock	Available	Available	Available
	MNCH program	Stock out since two months	Stock out since two months	Stock out since two months
	Civil Hospital Sukkur	Available	Stock out since 4 months	Available
	THQ Rohri	Available	Available	Stock out since 4 months (delay from manufacturer)
	RHC Bachal Shah Miani	Available (Before two months stock was out for 2 months)	Stock out (second line of therapy is in use that is Ampicillin + cloxacillin)	Available
	BHU New pind - PPHI	Available	Available	Available
	BHU Thikrato	Available	Available	Available
Kashmore @ Kandhkot	EDOH	Available	Stock out	Available
	THQ Hospital Kandkot	Available	Stock out	Stock out (one month)
	RHC Ghouspur	Available	Stock out since one month The Health Facility is with Integrated Health System (IHS) and MS told that HIS is struggling to provide medicine due to delay in timely release of budget	Stock out since 4 months The Health Facility is with Integrated Health System (IHS) and MS told that HIS is struggling to provide medicine due to delay in timely
	BHU Dari	Available	Available	Available
	BHU Khewaly	Available	Available	Available
Tando Muhammad Khan	EDOH	Available	Available	Available
	DHQH TMK	Available	Available	Available
	THQH TMK	Available	Available	Available
	RHC Rajo Nazamani	Available	Available	Available
	BHU Saeed pur Takara	Available	Available	Available
	BHU Sheikh Bhirkyo	Available	Available	Available
Tharparkar / Mithi	EDOH	Available	Available	Available
	DHQ Hospital	Available	Available	Available
	THQ Chachro	Available	Available	Available
	RHC Islamkot	Available	Available	Available
	BHU Kot Arbab Mir Muhammad	Available	Available	Available
	BHU Maln Hor Vena	Available	Available	Available
Karachi / Malir	EDOH	Available	Available	Stock out Since 5 months
	MNCH (Vertical Health Programme)	Stock out	Stock out	Stock out
	RHC Old Thanu	Available	Available	Available

District	Health Facility	ORS	Amoxicillin syrup	Zinc Syrup
	Sindh Government Hospital Murad Memon Goth	Available	Available	Available
	BHU Jaffer e teyyar	Available	Functional stock out (Physically present but medicines requisition submitted to government with zero stock in-hand report)	Stock out since 3 months
Shaheed Benazirabad (This district is without PPHI and all BHUs are run by DoH)	EDOH	Available	Available	Available
	DHQ hospital	Available	Available	Available
	THQ Hospital Sarkand	Available	Available	Available
	RHC Daur	Available	Available	Available
	BHU Karam Ali Jamali	Available	Available	Available
	BHU Gabhar Lakhair	Available	Available	Available

Table 18: Punjab Province - Availability of Stocks in Districts visited during the Field Study - June 2017

District	Health Provider	ORS	Amoxicillin syrup	Zinc Syrup
Rajampur	District Warehouse	Available	Available	Available
	DHQ	Available	Available	Available
	THQ Jampur	Available	Available	Available
	RHC Muhammad pur Dewan	Available	Available	Available
	BHU Kotla Eisa	Available	Available	Insufficient stock
	BHU Saleem Abad	available	Stock currently present but during one year period 6 days stock was out	Available
	IRMNCH & NP	Sufficient stock Not available, LHS of Saleem-abad has reported that since last two months Amoxicillin, ORS, and Zinc is out of stock		
Muzaffargarh	District Warehouse	available	available (Amoxicillin 250 mg/ 5ml was procured which need more dose accuracy for patient	available
	DHQ	Available	Available	Available
	THQ Kot adu	Available	Available	Available
	RHC sinawan	Available	Available	Available
	BHU jaday wala	Available	Available	Available
	BHU Muhammad Kot	Available	Available	Available
	IRMNCH & NP	Available (stocks for three months has been issued through the courier) no demand has been calculated from the field	Available	Available
Bahawalnagar	District Warehouse	Available	Available	Available
	DHQ	Available	Available	Available
	THQ Chistian	Available	stock out	Available
	RHC Madrisa	During last one year stock out of ORS occurred	Stock out of Amoxicillin syrup was observed which was solved on Jan 20117	Stock out occurred during last one year
	BHU 54/F	Available	Available	Available
	BHU 98/F	Available	stock out	Available
	IRMNCH & NP	Available	Available	Available
Pakpattan	District Warehouse	Currently available but stock out situation observed over the past one year		
	DHQ	Available	Available	Available
	RHC Bunga Hayat	Available	Available	Available
	BHU 28/SP	Available	Available	Available
	IRMNCH	Insufficient stock	Insufficient stock	Insufficient stock